

High Performance Computing for Weather and Climate (HPC4WC)

Content: Administrative

Lecturers: Oliver Fuhrer, Max Harvey, Simon Adamov

Block course 701-1270-00L

Summer 2026



Nice to meet you!



Oliver Fuhrer
oliver.fuhrer@meteoswiss.ch

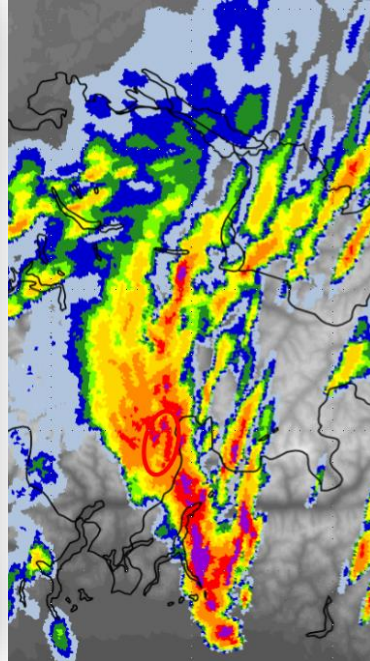


Simon Adamov
simon.adamov@env.ethz.ch

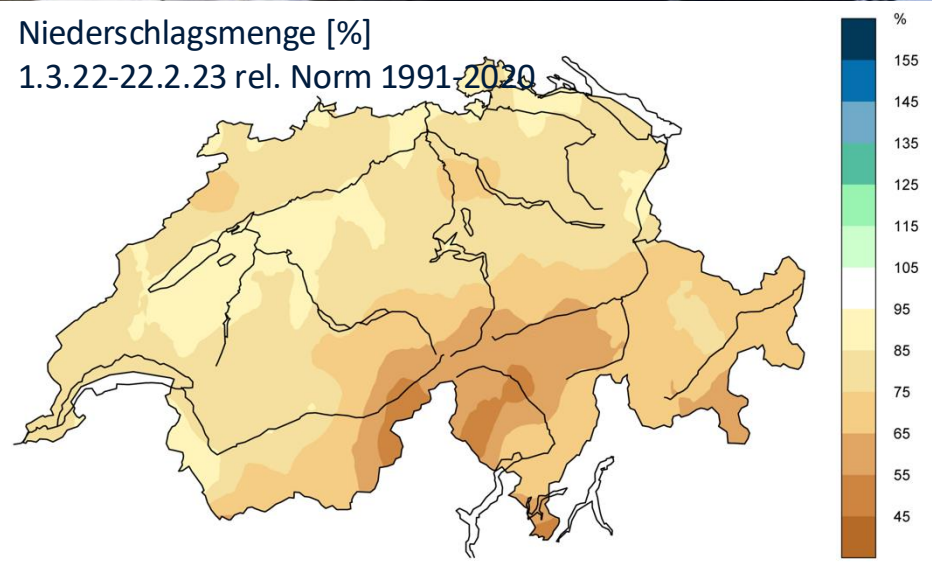


Max Harvey
max.harvey@env.ethz.ch

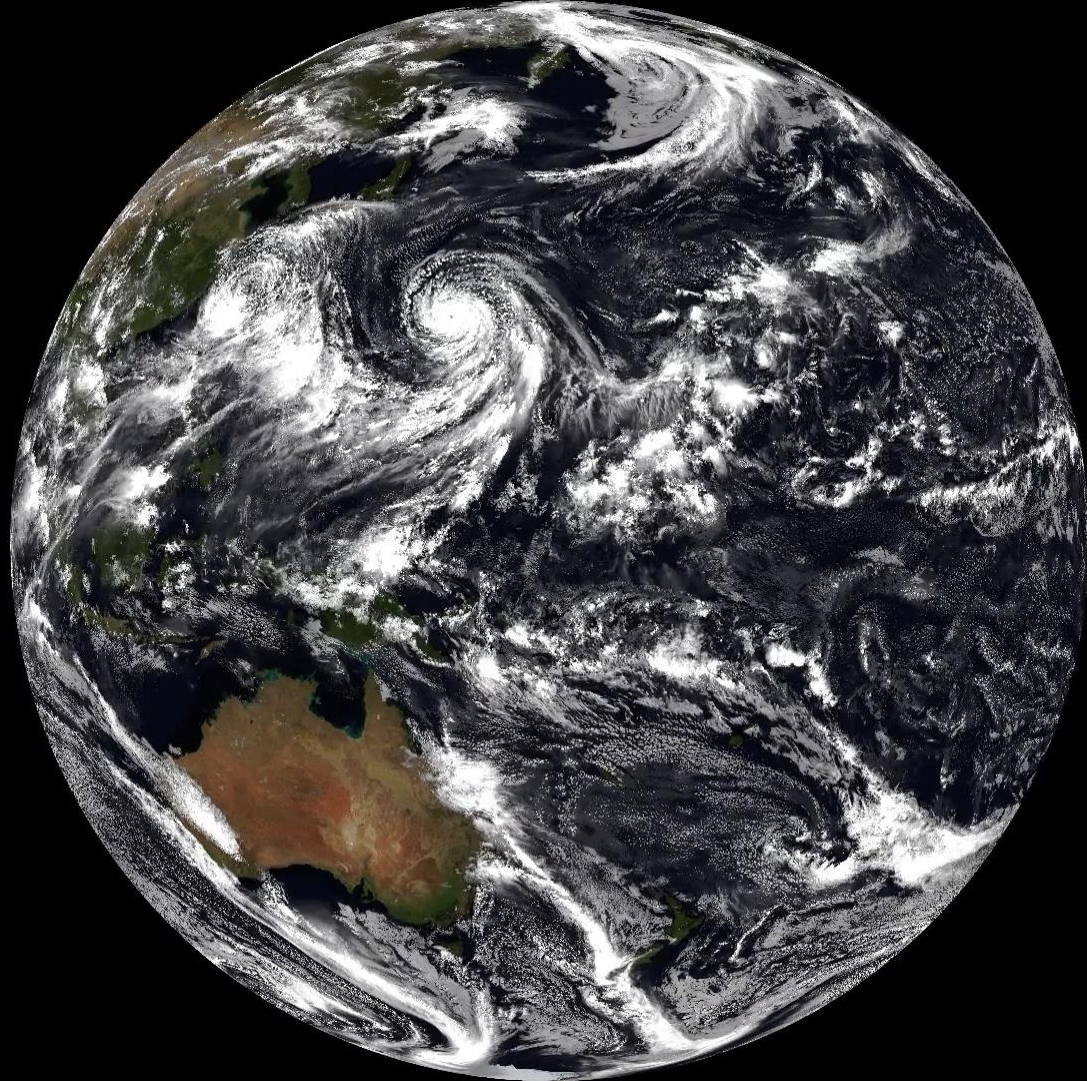
June 21, 2024, Misox, 8 fatalities



What's behind the warnings for such extreme events?



How do we know if this will be the «norm» in the future?



2016-08-11 18:00Z
258 Forecast Hours
FV3 3km

Weather and climate models run on supercomputers



Goals of Course

- Understand high performance computing concepts relevant for weather and climate simulations
- Able to work with weather and climate simulation codes that run on large supercomputers

Approach

“ I *hear*, and I forget. ”
I *see*, and I remember.
I *do*, and I understand.

(Chinese proverb)

- Lectures that explain concepts and give context (*hear*).
- Demonstrations of the concepts being applied (*see*).
- Practical exercises and a work project (*do*).

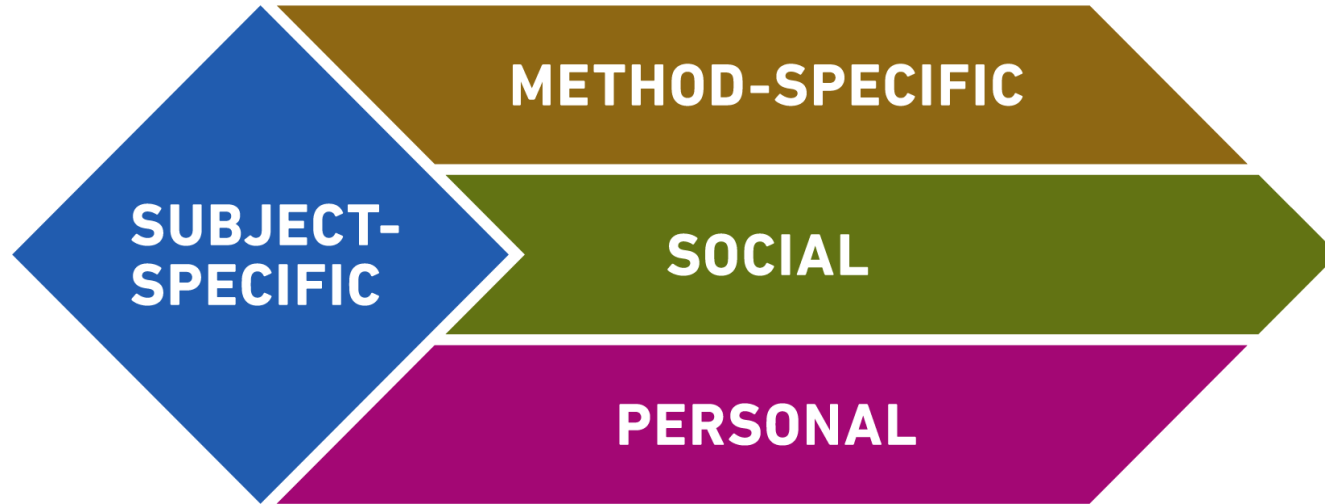
Questions, please!

ASK QUESTIONS - BY JAKEPOSEY

WWW.TOONDOO.COM



Competencies



Schedule

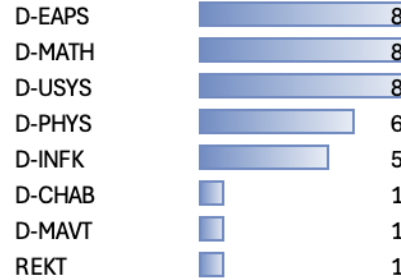
Monday	Motivation, stencil computations, memory hierarchy, lab environment	08:15 – 12:00	Morning session
Tuesday	Shared memory parallelism, OpenMP, performance metrics	12:00 – 13:30	<i>Lunch break</i>
Wednesday	Distributed memory parallelism, domain-decomposition and halo-updates	13:30 – 17:00	Afternoon session
Thursday	Hardware trends in supercomputing, GPU computing		
Friday	High-level programming, domain-specific languages, wrapup		

Currently registered students (eDoz)

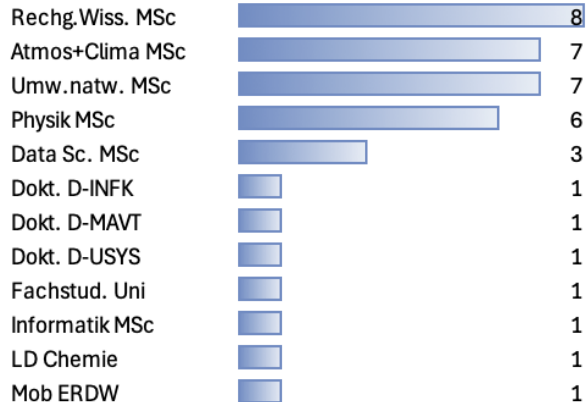
We are currently
33 registered
participants.

At the upper limit
for this course.

Departement



Studiengang



Prerequisites

- **Fundamentals of numerical analysis and atmospheric modeling**
 - Basic partial differential calculus and finite difference methods.
 - e.g. ETH course “[Numerical methods in environmental physics](#)” or “[Weather and climate models](#)”
- **Experience in a programming language (C/C++, Fortran, Python, ...)**
 - We will read and write [Fortran](#), C++ and [Python](#) in this course.
- **Experience using command line interfaces in *nix environments (e.g., Unix, Linux)**
 - Familiar with work in the [command line shell](#) and the most commonly used shell commands.
 - Can logon to Linux system via ssh and can work remotely on that system.
 - We will work on the [Alps supercomputer](#) at the [Swiss National Supercomputing Center \(CSCS\)](#) in Lugano in this course.

If you think this course might not be suitable for you, talk to us!

Quick Poll



Linux / Unix, Terminal



Programming (Python, Fortran, C/C++, ...), Compilation, Debugger, git



Jupyter Notebooks



PDE, Stencil, Weather Model



Numpy, OpenMP, MPI, CUDA

Practicalities

- **All course material on GitHub repository** (slides, notebooks, codes, ...)
<https://github.com/ofuhrer/HPC4WC/>
- **Questions related to course and projects in dedicated Slack workspace**
https://join.slack.com/t/hpc4wc2026/shared_invite/zt-40ch90885-cjsw974wAFB~lhM8qGSzyw
- Generally, try to use public channels for questions since others probably have the same questions.
- **Lectures are not recorded**



How to earn credits (3 ECTS)

- **Attend the block course** (and participate actively!)
- **Work project**
 - Choose group and topic
 - Hand in working source code and report (max. 10 pages)
 - Projects will be graded
 - **Deadline: 31. August 2026**
- Credits are awarded if course attended and grade of work project ≥ 4.0
- Same rules apply for BSc, MSc, and PhD students

Work project



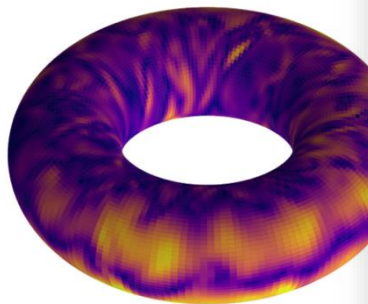
- **Work in groups of 3-4** (no individual projects)
 - Programming is not a solitary art!
- **Topics [Google Docs](#)**
 - Have an idea of your own? Cool, discuss with us to shape it into a project!
 - Each project must have a software development and performance evaluation part and has to be related to course material
- **Grading**
 - 25% correctness (compiles & runs, results correct, no bugs)
 - 25% quality (structure, clean code, comments, naming, tests, error handling)
 - 25% performance (depending on work project)
 - 25% report (maximum 10 pages)
- See [last year's projects](#) for examples

Examples

Shallow water equations on a toroidal planet
in the domain-specific language GT4Py

Killian P. Brennan, Dana Grund, Joren Janzing, and Franco Lee
Course Project: High-Performance Computing for Weather and Climate

August 2023

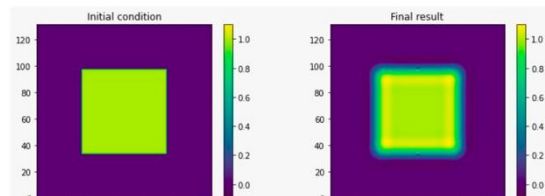


High Performance Computing for Weather and Climate, FS 2022

Work Project: Communication strategies for halo updates

Submitted by:

Victoria Bauer, Sarina Danioth & Julia Dworzak



CACHE HIERARCHY AND LOOP OPTIMIZATIONS

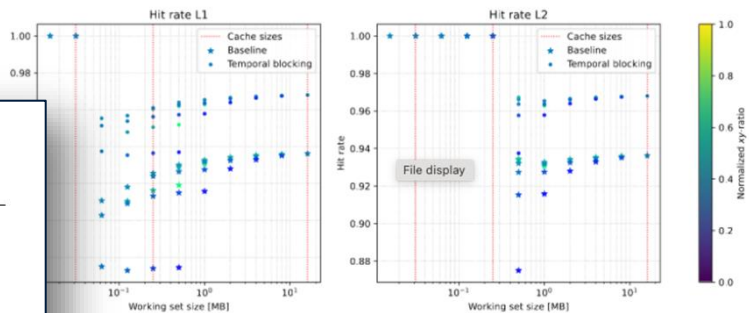
Report

Basil Ruch*

ETH Zurich, Switzerland
Department of Mathematics

David Strassmann†

ETH Zurich, Switzerland
Department of Mathematics



4: Recorded hit rates in the L1 (left) and L2 (right) cache for the 2D stencil. Star marks represent the baseline and circles the temporal blocking algorithm. The color indicates the ratio of the two dimensions: a value of 0.5 means that x and y dimension are equal.

Use of AI-assistants (e.g. Claude, Codex, ChatGPT)



What's Allowed

- Understand code
- Debug errors or explore fixes
- Ask conceptual or performance-related questions



What's Not Allowed

- Blindly copy AI-generated code / answers without reviewing and understanding
- Skip required learning steps or performance validation

What did we use AI for?

To update course to new Alps environment

To develop and debug some bonus questions

To check information about GH200 hardware

Not to generate synopsis and course content in general

Not to review/grade your projects

Lab exercises

- Swiss National Supercomputing Centre <https://www.cscs.ch/>
- Alps supercomputer (#8 on [list of 500 largest supercomputers](#) worldwide)



CSCS Accounts

- Everybody has a unique user name (course_XXXXX) and password.
- **Do not share you login / pwd with anybody else.** Accounts with suspicious activities will be closed down by CSCS immediately.
- We have a shared quota of ~3'000 node hours for using the CSCS supercomputers for this block course.
 - Do not launch jobs with more than 1 node without checking with us first.
 - Do not leave your JupyterHub Server running if you don't need it.
- **Do not contact CSCS first** if you have trouble. Ask us or use the Slack workspace to get your issues resolved.
- Take a look at the [CSCS Code of Conduct](#)

JupyterHub

- Lab exercises will all be conducted on <https://jupyter-santis.cscs.ch/>, the JupyterHub portal of CSCS.
- Interactive development and computing environment.
- If things get stuck or go wrong, it's always possible to “Stop Server” and “Launch Server” again.
- Jupyter notebooks auto-save and *almost* certainly no work will be lost.

Server Options CSCS

GPUs

1

Account

hpc4wc-
course2026-ethz

Job Duration

8 hours

Advanced options

Custom uenv

prgenv-gnu/26.3:v1

Overrides the default uenv

Uenv view

default

Custom Reservation

hpc4wc-course2026

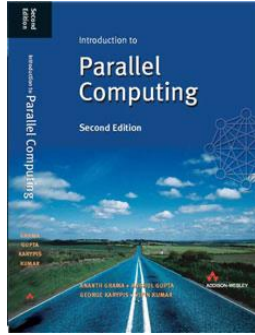
Optional custom reservation

Start

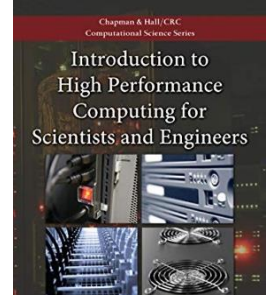
We rely on a bleeding-edge supercomputer!



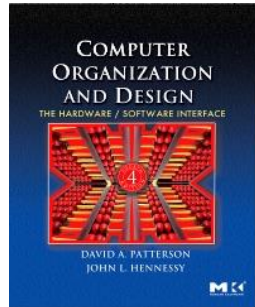
Literature & Links



Introduction to High Performance Computing for Scientists and Engineers, G. Hager and G. Wellein, CRC Press, 2011
(available online at [ETH](#))



Parallel Computing, A. Grama, A. Gupta, G. Karypis, V. Kumar
(available free online)



Parallel Programming in MPI and OpenMP, V. Eijkhout
([Link to course](#))



Computer Organization and Design, D.H. Patterson and J.L. Hennessy (available online at [ETH](#))